A Proposed Machine Learning Based Approach to Support Students with Learning Difficulties in The Post-Pandemic Norm

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***Abstract*—Over the years, there have been many factors that have had an influence on the landscape of higher education in the UK. These factors include the rise of tuition fees, the introduction of the teaching excellence framework and the formation of office for students. A key performance indicator that has an impact on these factors is student experience, which is influenced by positive or negative feedback and engagement. Although this forms a key part of the learning environment, it is still perceived as one of the weakest aspects when it comes to enhancing the student experience especially for the students with learning difficulties. During the recent pandemic, significant levels of changes have been introduced to the teaching and learning approaches. Machine learning approaches are proven useful for providing flexible solutions for various problems in different fields. With the focus on the students with learning difficulties; this paper proposed a machine learning based approach to support such students and analyse the complexities of their learning difficulties to make sure they benefit from the new approaches in the post-pandemic era. The proposed approach has been tested initially for dyslexia, where the main complexities such as recognition of words, fluency in reading and writing are analysed. This research will lead to the novel introduction of intelligent approach to revolutionize the learning ability and overcome any learning difficulties at different learning and teaching levels.**

Keywords— Learning and Teaching, Machine Learning, Learning Difficulties, Students, Post-Pandemic.

I. INTRODUCTION

Learning difficulties such as dyslexia, dysgraphia, dyspraxia, and many others interfere with the academic achievements of the students, and it creates some of the long-term consequences in their non-academic activities. It was stated that about 5 - 10 % of the world's population are affected by learning difficulties/ disorders. The learning difficulties can be predicted using various medical tests. The human experts will be scoring those tests and will decide whether those students need a particular strategy for their education (the strategy determined based on the achieved score). These assessments are costly, painful and time consuming. The learning disorders can be investigated from many viewpoints like the prevalence that is depends on the gender and the dependence on the family and the alternative definitions of the learning difficulties. The fact that can be trusted is that the identification of the dyslexia issue is possible with the help of high reliability even when the correct nature of the dyslexia is not known. This is the valid disorder when compared to the other disorders. There are many symptoms for dyslexia, such as the ability of individuals to read. Dysgraphia is another learning disorder that impacts the ability of the individual to write, and these disorders will remain with the individual for long time over the age and this can be mitigated with appropriately designed sessions. Without the proper support, these kinds of disorders can lead to an academic failure. But by providing the appropriate education strategy, such students can learn and become successful. Sometimes the students get the support in different forms, e.g., special teaching lessons, extra time for the exams and the tuition for them and dedicated academic staff who will be helping them in the classroom (through someone to one sessions). In order to get the support, the certificate must be provided by the specialists who will be taking in charge of that child. The concept behind this learning model is to connect education with technology to meet the different needs of each student [1, 2].

In today's academic climate, being able to effectively engage students is seen as a challenge, where the consequences can have a negative impact on student experience. This is something that higher education institutions are paying a close attention to, as they are actively moving away from traditional methods and trying different ways to engage and motivate students. With any of the new methods, the challenge lies in constructing educational experiences that do more to trigger the interest of the student. One of the ways of achieving this is to increase enjoyment and motivation levels within the student's learning activities and milestones [3-5]. Active learning methods are considered more effective than traditional methods for engaging students who are disengaged. As these methods promote positive learning experiences and student motivation is improved through achievement. One way to engage students is to facilitate a learning paradigm that promotes interaction and collaboration between students and academics. This in turn cultivates a sense of belonging and co-operation within the student, which can have a positive impact in student engagement. The work in this paper builds on this concept, as

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we present the implementation of an innovative, holistic teaching & learning framework for Computer Science based final year projects [4, 6].

Machine learning is a technology that can be used to help the students affected with learning difficulties and the students with neural disorder. It helps the students to personalize and predict the different learning levels using the web mining technology for example. Where the pre-academic skills can be used with the students to avoid difficulties with the learning approaches. Machine learning is often used with the career choice that is organized with the learner to identify the read, write and spelling difficulty with the identification of the disability with the child. Learning difficulties are caused due to neurological disabilities that is accessed and specified with the visual and hearing disability. There are different algorithms identified with the prediction of disabilities that are used with the machine learning techniques that is applied with the predictive modelling and descriptive modelling. As there is only a limited number of pathologists, then manual process will be significantly time consuming. Machine learning opens new ways to help students with disabilities. Students with autism which is a neurological disorder need a personalized development system for their daily activities. Technology can play a substantial part in that as well; where the system includes the ability to predict the learning level of the user, and generating multi-modal learning materials [7, 8].

The proposed machine learning based approach can widely support the identification process and provide an accurate analysis for the different level of the learning difficulties and disorders. This research will aim to build a comprehensive multimodal database to support the development and introduction of such approach. This will lead to the interest that is growing in the development of the tools that is capable of the extraction of the knowledge data in an automatic way. Students with a disability often need subject matter presented through different methods, therefore it is imperative that these technological advances benefit all students with different learning styles. The proposed machine learning based approach will be able to analyse the difference between the students with learning difficulties and the students without such difficulties and will build an effective model to suggest the appropriate support and strategy which can be personalised based on each individual case. Then the model will be trained on the features that are arrived by the process of analysis, the images and the audio recordings to monitor the level of each individual and the improvement and engagement level as well.

II. Literture Review

A. Student Engagement

The effective lecture should always have two types of activities taking place; teaching and learning. The teaching activity is associated with introducing and explaining the new topic with all the relevant information and examples. The explanation process includes a full discussion about what has been said with the students; this discussion helps the students filling any gap related to the topic and gives the lecturer a good indication about the students learning progress. The students are encouraged to demonstrate their leaning and engagement through the discussion, questioning and effective interaction with the lecturer to achieve learning activities with instructional scaffolding [9]. Student engagement can be maintained and improved through the class discussion, working closely with the students, motivate the group working, encourage class and tutorial debate, and placing the student at the heart of teaching material evaluation [10].

The idea of student engagement has been defined by different researchers using different terms, including academic engagement, student engagement, learner engagement and school engagement [11]. The employment of technology to deliver teaching contents, connect students, and provide the ability to learn at any place and anytime is increasing. However, the main challenge here is to maintain the students' engagement. The work in [12], has reviewed and examined the available techniques to measure the students' engagement in technology-mediated learning. These measures need to be useful and adaptable to the learning challenges based on technology. This research work identified strengths and limitations of the measurement of student engagement, which will lead to the effective approach to measure the student engagement especially in the technology-mediated learning environment.

Currently, many researchers are investigating best practices to deploy people and technology to engage students in an efficient learning environment. There is a range of indicators deployed in research study [12] to measure the engagement. These indicators include behaviour, cognitive and emotional one. Where the emotional engagement indicators involve positive or negative emotions towards different factors such as learning, classmates, or instructors, the social interactions and a sense of community can be also included under this indicator. Moreover, the cognitive engagement includes cognitive strategy use, beliefs, values, and planning, however these may not always be visible (particularly from external point of view) and may need a self-reporting. Finally, the behavioural indicator which can include attendance, participation, coursework completed, time logged in, and any other behaviours related task. In a technology-based learning environment, the behavioural engagement can be evaluated based on a computer measure factor such as the completion of coursework, the frequency of logins to a certain website, responses, number and frequency of postings, expressing views, the number of podcasts, online resources and how much time is spent using or exploring these resources.

In a technology-based learning environment there are different ways to measure student engagement, including quantitative self-report, qualitative measures, quantitative observational measures, using the indicator of frequency, and other measuring approaches such as some studies are arguing that a high learners' performance or high completion rates can give evidence of student engagement [12-14].

The work in [15] presented a collaborative learning study to evaluate the student learning. The achieved results pointed that deployment of mobile technology is combined with positive students' attitudes and the critical thinking level of them is more closely engaged with the tools used than with the style of learning settings. As the student's responses on a mobile device showed significantly less critical thinking than those who used a computer keyboard or wrote responses by hand.

From the students' point of view, the students themselves became aware of the importance of engagement, as revealed by the student academic experience survey. They showed that the main reason for their experience not matching their

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expectations is because the students did not place enough effort themselves [16]. The undergraduate students' research projects give the students the opportunity to demonstrate the development of their own working and presentation skills, knowledge, and the ability to perform an independent study and initiate their own topic/field. They can also employ their own way of working and researching that is aligned with their career goals, personal interests and expand their knowledge. There are different properties for the final-year project; for example: the project should be an extended piece of work/ research/ inquiry-based, it should be related to a discipline or has multidiscipline points, it should demonstrate a good knowledge of the field and supported by wide range of resources. The project should demonstrate a critical thinking and evaluation for the problem in question with a clear contribution and has the ability to demonstrate the outcomes in an appropriate way [17].

B. Unstructured Learning / Independent Research

The independent directed learning can be defined as learning in which students are directed by the contents of the curriculum and supported by academic members of staff and the learning environment in general. In this learning the students play an active role in their learning experience and improving it. For this type of independent learning there is an explicit structure and ongoing support for students transitioning from level 3 to a higher level, where a complete independent learning is expected from the student especially in the final year. This learning type provides flexibility and inclusivity especially for the increased use of learning technology and online learning which can meet all the different learning preferences for the students and improve their engagement [18].

The research study performed by [19] found that students do not have a clear meaning of independent learning, what is its purpose, its difference from the learning at college or school and its important role in the preparation for student's employment. In most cases students understand the independent learning through trial and error not through the right disciplinary community. It was also found that the students from different disciplines focus their efforts on the duties that contributed directly to the assessment. This study indicates that on average, 13 hours per week are spent on independent learning; where this number increases to 16 when studying hard-pure subjects and goes down to 10 when studying soft-applied. Moreover, the first-year students spend less on independent learning than second- or third-year students, and the females spend 4 hours more than the males. It is indicated that if the students know what to do then they intend to do more independent learning in an easy approach, this also increases the group work and the collaboration between the students. The research shows that the students tend to discuss different personal issues to facilitate their independent learning, however, when they talk about their own independent learning experience, they concentrate on the assessment and exams. The students indicated that preparation at school or college level and performing some workshop can help them improve their efficient independent learning skills [18, 19].

Another research study pointed to the importance of using formal and informal social networks to improve the independent learning skills in higher education, which enables the students to discuss, evaluate and share their views and knowledge with other peers [20]. On another note, students' motivation can be assured through different aspects such as the emphasis on transferrable skills (interview skills, presentations, writing skills, critical analysis) and the employability issue on one hand. On the other hand, the lecturer will have a more enjoyable session with the high spirit motivated students. It is also important to fully understand the needs for our students and why they are coming to the classroom and encourage them to participate in the class discussion and avoid "I do not know" statement. Students can present their work in group or in front of the whole class, this practice improves their communications skills and gives the rest of the students the opportunity to learn and communicate effectively from their peers [21].

C. Use of Technology in Teaching

It is important for learning in technology-based environment to have flexible thinking, which can serve as a key competency to address challenges in the educational system. The flexible thinking includes acceptance of technologies, open-mindedness to the ideas from others, and the ability to adapt to changes in learning environment [22]. The self-regulate learning process which includes determining the learning goals and setting up the methods and materials to reach these goals has been assessed in the research study [23]. Moreover, self-assessment and peer-assessment have been deployed as a learning procedure [24].

In study conducted by [25], the effects of computer science education have been determined where the results show that fostering the quality of computer science education is very important beside the right motivation for the students to pursue science, technology, engineering, and mathematics studies. Another research study focused on the flexible thinking as a key important competency for learning in technology-enhanced environments. It has generated a scale for the flexible thinking in learning, which can help promote the open-mindedness for other people ideas, accepting technology changes and adapting these changes into the learning environment. It was suggested that the proposed scale can be used as a research tool and a self-evaluation tool [22]. The results achieved by [26] can be employed to create an effective online system which has the elements for better strategies in the students' learning. The technology employment in the higher education sector can have a significant impact, due to different recent challenges, such as an increase in student numbers, a limited public fund and increased demand from students to have a high-quality learning experience.

In the computing discipline we share some of the concerns highlighted in the HEA report, such as communication, employability, low retention therefore we have proposed some interesting activities at the beginning of the term to make sure that the students are engaging with the university environment, knowing each other, and communicating effectively with the academics at different levels. In teaching approach there are two main approaches student-centered approach and teacher-centered approach. However, in computing discipline area the teacher-centered approach is dominantly used, where the lecturer has expertise and the knowledge of the proposed area, and sometimes the students monitor and adapt the method used by the lecturer [27]. Hence introducing intelligent approach to support the learning difficulties faced by significant number of students can take the existing teaching and learning approaches to a better and more comprehensive achievable level.

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III. Learning Difficulties- Dyslexia Case Study

There are many types of learning difficulties and hence the variety of the available assessment tests. The age of the student and the problem type will determine the required test to indicate and measure the level of the learning difficulty. The overall evaluation will start with the testing as well as the physical examination in order to rule out any hearing or visual impairment. Professional experts can be involved in the process of testing. The evaluation purpose for the learning disability is to find the strength and the weakness of the student and to understand the difficulty that is faced. The data that is gained from the evaluation will be critical for identifying how the authorities of the school or the parents will give the good learning environment for that child [28]. We will be using this data as part of the developed model. As there are significant number of assessment lists (including signs and symptoms) for the learning difficulties.

Dyslexia is considered as the commonly referred issue in the learning difficulties. There was significant level of misunderstood terms related to dyslexia which are confusing many people. Identifying the individuals as dyslexic will support them to understand the experience, they have but this will not give any data or the direction regarding how to help and teach a particular person on how to overcome the writing and reading difficulties. Because of this, the term dyslexia was avoided in the field of educational contexts with the priority given in the term of learning difficulty. Although there was a continuous utilization of the term dyslexia in the research, but many members in the public are not fully aware of it. Moreover, the works on dyslexia showed that there is no agreement on the meaning of this term, dyslexia, across many countries that speak English. This became as a widely accepted fact that dyslexia is a form of learning difficulty and it also has the biological traits, and it is different from the other types of learning difficulties. Some definition of dyslexia involves the phonological deficit. In the last decade the meaning of dyslexia was moved from the IQ in the definition and accepted that this phonological deficit must be included in the definition of dyslexia. The researchers on the term dyslexia were sure that there is no deal on the effects and the causes of the dyslexia [29, 30].

The students with dyslexia can have trouble in the key skills like writing, spelling and words pronunciation. It can affect the individuals all over their life and its impact might be modified at different stages of the individual's life. This will lead to very tedious process for the students to achieve in the academic area in the instructional environment [31]. The main cause as per the researchers of dyslexia was not clear but the studies of brain imagery will display the differences in the development and the functions of the dyslexic person. Many people with dyslexia might found that they have the issues in finding the isolated speech sounds in the word or in the learning how the letters will represent the sounds and the factors in the difficulties of their reading. The dyslexia is not because of the lack in intelligence nor the desire in learning process. By employing the correct method of teaching, the person with the dyslexia can learn in an efficient and successful manner.

Dyslexia originates from the neurological parts. This is categorized by the difficulties in the accuracy as well as the fluency in the word recognition and in the poor spelling as well as the decoding abilities. These issues will result in the phonological component which is unexpected. Some issues include the problems like reading comprehension, reading experience that will affect the growth of vocabulary as well as the background knowledge. The impact of dyslexia differs from person to another which is depending on the severity of the conditions and the effects of remediation. The main issues are the recognition of words and the fluency in reading, writing and in spelling. Some dyslexics have the ability to manage to learn the spelling task and the early reading particularly with the good instructions, but the most important problems are that the grammars, essays writing and the understanding of the textbook materials. In addition to the issue of the spoken languages. These kinds of difficulties are tedious to find but they lead to many important issues in the schools and workplaces. They might affect the self-confidence of the student and being the least capable. In some cases, experiencing the stress due to the academic problems can discourage the students from continuing their education [32, 33].

There are different processes followed by organisations /local authorities to measure the learning difficulties. For example, in the response to intervention, which is used to find the student with learning difficulties, the students will be provided with the acquisition of issues in early skills of literacy by the intensive instructions of reading. If the learning of the students does not get the required grades, then other kind of disorders will be identified in the process of reading. Many students are found as dyslexic and hence be qualified for the special education process. Schools are asked to find the students with the learning disabilities in the earlier stage [34]. This evaluation was traditionally included in the achievement of testing and the assessment of underlying skills. This includes the listening skills and the language skills, skills of phonological that include the awareness and the ability of the students to name the letters and the names. Beside the ability of the students to read the list of words is the isolation and the words in the context. If the profile is with the characteristic of dyslexic, then the plan should be developed that will include the correct accommodations. The process of testing will be done by the trained school or some other specialists. Hence the proposed approach based on machine learning can be used for the analysis and assessment level of such learning difficulty. It can also be used to classify the categories and the development level of dyslexia.

IV. The Proposed Approach

Fig. 1 illustrates the key components of the proposed approach based on machine learning to support students with learning difficulties and identify the level of these difficulties. Two types of data will be acquired, quantitative and qualitative data, psychologists' experts will be joining the team during the data acquisition stage.

- The quantitative data set will include video recording to capture the face gesture details, eye and lips movements. Electroencephalogram (EEG) data: EEG is used to rule out various health-related conditions like seizure, epilepsy, encephalitis, brain tumor, memory problem, stress, sleep disorders, stroke, dementia, etc. Moreover, the body temperature, heart rate, blood pressure.
- The qualitative data: the expert psychologists will provide the participants with different / multi activities to examine the behaviours of the participants while performing the writing, reading

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and memory related activities. The designed questions will be asked at several stages.

The second stage after collecting the data is the pre-processing stage, appropriate features extraction, identifying the level of the required assessment. Once we have all the required data available, a dynamic machine learning model will be trained to identify the learning difficulties, their level, and recommend a personalised strategy for each individual based on the available information and predicted ones. The identified and predicted strategy will be fed to, connected to the learning, and teaching approach in the organisation in question, to keep real time connection with them for continuous feedback and a step by step monitored implementation of the learning strategy. Based on the severity case of each individual the regular monitoring and feedback to the ML approach can be set up at weekly or monthly basis. A significant level of multi organisations collaboration is required, beside suitable infrastructure is needed to perform and maintain an appropriate implementation of this approach. Regular report will be generated based on the obtained data sets, organisation feedback, subjective feedback, and objective feedback.



Fig. 1. The Proposed Machine Learning Based Model To Support Student with Learning Difficulties.

As each individual with learning difficulties requires a significant amount of time and effort to identify them and their levels, hence the proposed approach will provide a novel platform for the identification of such difficulties and determining the level of these difficulties to support both the learning and teaching organisations as well as the students.

A machine learning based approach can be widely effective in recognising different patterns in the reading, typing, writing, web-based games, handwriting; where such recognition needs a significant amount of time when done by human and will be subject to human error. Moreover, such approach will be effectively in analysing multiple types of data such as text, MRI scans, eye movement, EEG scan and the image. The proposed approach can be trained to recognised different native languages such as Spanish, Swedish, French, Dutch, Greek, Malay, English, polish, German, and others. The acquired data will go through robust pre-processing stage to identify different representing features, filter/pre-process the voice and movement images (e.g., high and low pass filters, image domain conversion e.g., different wavelet/frequency transforms) and convert the other types of data into a compatible and coherent data format. The pre-processing can help in identifying the related parameters and avoiding or removing the null values. This step is essential to prepare the right data sets for training the machine learning based approach. Once the pre-processing and the feature extraction are completed the relevant features will be identified and the range of values will be assigned within the model. The next process is to find the set of features that is dominant and important for finding the class for the object. Once the features are successfully selected, then the process of designing and training the appropriate machine learning based approach are conducted comprehensively utilising all types of data. These data sets will be employed for the training, testing and the validation parts. Overall performance evaluation will be performed afterward to make sure all learning strategies are included and the right feedback is put in place to fully support the students with the learning difficulties.

V. Conclusion and Future Work in Post Pandemic Landscape

This research work has proposed a machine learning based approach to support students with learning difficulties and identify the level of these difficulties. Such approach will facilitate the key and fundamental strategy's change to the learning and teaching approach for the students with the learning difficulties. This research will lead to a novel introduction of intelligent approach to revolutionize the learning ability and overcome any learning difficulty at different learning and teaching levels in the post-pandemic era. It will create an innovative generic smart knowledgebase school care coordination model, which adopts a comprehensive strategy to bring education, health, and social care services into a single live application to meet the needs and address all the learning difficulties. The future steps will include the acquisition of ethical approval, the data collection (quantitative and qualitative data), education reports for the participants, progress report (for their primary/secondary studies). All the participants will be fully informed about the study and the privacy issues will be dealt with comprehensively through the right ethical procedure. This research approach will form the foundation of further research into the combination of the education approach, health, and social care services for students with learning difficulties. It will form the info-structure to model and implement a generic

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novel approach which involves a significant level of multi organisations collaboration. Such collaboration includes the schools, universities, local authorities, private and public companies. A significant amount of time and huge amount of effort will be saved through this approach. The developed approach will be employed to coordinate, integrate, and monitor education, health, and social care activities between different actors (formal, informal and students) involved in the schools and universities care network. Moreover, the proposed approach will be capable of providing a personalised learning strategy for each individual with learning difficulties. It will employ quantitative and qualitative data sets with an objective and subjective evaluation of the outputs to widely support the students. The achieved results will be generalised and passed on to the participating research partners and local authorities to benefit from these results and applying them within the learning strategy.

REFERENCES

[1] Hussain, M., Zhu, W., Zhang, W., Abidi, S.M.R.. and Ali, S. Using machine learning to predict student difficulties from learning session data. Artificial Intelligence Review, 52(1), pp.381-407, 2019.

[2] Mounica, R. 0., Soumya, V., Krovvidi, S., Chandrika, K. S., & Gayathri, R. A Multi Layer Ensemble Learning Framework for Learning Disability Detection in School-Aged Children, 2019.

[3] M. Kangas, P. Siklander, J. Randolph, and H. Ruokamo. Teachers' engagement and students' satisfaction with a playful learning environment. Teaching and Teacher Education, 63:274–284, 2017.

[4] P. Siklander, M. Kangas, S. Ruhalahti, and S. Korva. Exploring triggers for arousing interest in the online learning. In International Technology, Education and Development Conference, pages 9081–9089, 2017.

[5] K. A. Renninger and J. E. Bachrach. Studying triggers for interest and engagement using observational methods. Educational Psychologist, 50(1):58–69, 2015.

[6] S. Freeman, S. L. Eddy, M. McDonough, M. K. Smith, N. Okoroafor, H. Jordt, and M. P. Wenderoth. Active learning increases student performance in science, engineering, and mathematics. Proceedings of the National Academy of Sciences, 111(23):8410–8415, 2014.

[7] Ambili K, A. P. A Framework for Learning Disability Prediction in School Children Using aritficial Neural Networks, 2016.

[8] Jockim, J.M., K, M., & S, K. Reel or Real - Survey of Machine Learning to Adversarial Machine Learning Algorithms for Fake Content Detection in the Education Domain. SSRN Electronic Journal. https://doi.org/10.2139/ssrn.3735766, 2020.

[9] R. K. Sawyer. The Cambridge Handbook of the Learning Science. Cambridge University Press, 2006.

[10] Thomas Liz and Helen May. Inclusive learning and teaching in higher education. The Higher Education Academy, 2016.

[11] Reschly, A. L., & Christenson, S. L. Jingle, jangle, and conceptual haziness: evolution and future directions of the engagement construct. In S. L, 2012.

[12] Curtis R. Henrie, Lisa R. Halverson, Charles R. Graham, Measuring student engagement in technology-mediated learning: A review, Computers & Education, Volume 90, 1 December 2015, Pages 36-53.

[13] Liang, H.-N., & Sedig, K. (2010). Can interactive visualization tools engage and support pre-university students in exploring non-trivial mathematical concepts? Computers & Education, 54, 972e991

[14] Rowe, J. P., Shores, L. R., Mott, B. W., Lester, J. C., & Carolina, N. (2011). Integrating learning, problem solving, and engagement in narrative-centered learning environments. International Journal of Artificial Intelligence in Education, 21, 115e133.

[15] Houston Heflin, Jennifer Shewmaker, Jessica Nguyen, Impact of mobile technology on student attitudes, engagement, and learning, Computers & Education, Volume 107, April 2017, Pages 91-99, ISSN 0360-1315.

[16] Alex Buckley. UK Engagement Survey, The second pilot year. The Higher Education Academy, 2016.

[17] Mick Healey, Laura Lannin, Arran Stibbe and James Derounian. Developing and enhancing undergraduate final-year projects and dissertations. The Higher Education Academy, 2013.

[18] Thomas Liz, Robert Jones and Dr James Ottaway. Effective practice in the design of directed independent learning opportunities. The Higher Education Academy, 2015.

[19] Thomas Liz, Christine Hockings, James Ottaway and Robert Jones. Independent learning: student perspectives and experiences. The Higher Education Academy, 2015.

[20] Terry Mayes, Derek Morrison, Harvey Mellar, Peter Bullen and Martin Oliver. Transforming Higher Education Through Technology-Enhanced Learning. The Higher Education Academy, 2009.

[21] Michael F. McMahon. Employability, Transferable Skills and Student Motivation. The Higher Education Academy, 2007.

[22] Miri Barak, Ariella Levenberg, Flexible thinking in learning: An individual differences measure for learning in technology-enhanced environments, Computers & Education, Volume 99, August 2016, Pages 39-52, ISSN 0360-1315.

[23] Eugenia M.W. Ng, Fostering pre-service teachers' self-regulated learning through self- and peer assessment of wiki projects, Computers & Education, Volume 98, July 2016, Pages 180-191, ISSN 0360-1315.

[24] César Domínguez, Arturo Jaime, Ana Sánchez, José Miguel Blanco, Jónathan Heras, A comparative analysis of the consistency and difference among online self-, peer-, external- and instructor-assessments: The competitive effect, Computers in Human Behavior, Volume 60, July 2016, Pages 112-120, ISSN 0747-5632.

[25] Ahlam Lee, Determining the effects of computer science education at the secondary level on STEM major choices in postsecondary institutions in the United States, Computers & Education, Volume 88, October 2015, Pages 241-25 5, ISSN 0360-1315.

[26] Miran Zlatović, Igor Balaban, Dragutin Kermek, Using online assessments to stimulate learning strategies and achievement of learning goals, Computers & Education, Volume 91, 15 December 2015, Pages 32-45, ISSN 0360-1315.

[27] Andrea Abbas, Joan Abbas, Kira Brayman, John Brennan, Orkhon Gantogtokh. Teaching excellence in the disciplines. Higher Education Academy, 2016.

[28] Kayleigh K Hyde. Applications of Supervised Machine Learning in Autism Spectrum Disorder Research: a Review. https://linkspringer.com/article/10.1007/s40489- 019-00158-x I 1-19, 2019.

[29] Popov, M. &. I. T. Knowledge Model for Developing, Searching and Using Personalized Learning Content for Learners, Having Dyslexia Disability.. In Proceedings of the 21st International Conference on Computer Systems and Technologies' 20, pp. 258-265, 2020.

[30] Shahriar Kaisar. Developmental dyslexia detection using machine learning techniques : A survey. https:j/doi.org/10.1016/j.icte.2020. 05.006, 1-4, 2020.

[31] Kariyawasam, R. N. Deep learning based screening and intervention of dyslexia, dysgraphia and dyscalculia.. In14th Conference on Industrial and Information Systems, pp. 476-481, 2019.

[32] Nilashi, M., Ibrahim, 0., Ahmadi, H., Shahmoradi, L. and Farahmand, M., 2018. A hybrid intelligent system for the prediction of Parkinson's Disease progression using machine learning techniques. Biocybernetics and Biomedical Engineering, 38(1), pp.1-15, 2018.

[33] Spoon, K. C. D. &. S. K, 2019. Towards Detecting Dyslexia in children's handwriting using neural networks. In Proceedings of the International Conference on Machine learning Al for Social Good Workshop, long Beach, CA, USA, pp. 1-5, 2019.

[34] Sobnath D., Isiaq S.O., Rehman I.U., Nasralla M.M. Using Machine Learning Advances to Unravel Patterns in Subject Areas and Performances of University Students with Special Educational Needs and Disabilities (MALSEND): A Conceptual Approach. In: Yang XS., Sherratt S., Dey N., Joshi A. (eds) Fourth International Congress on Information and Communication Technology. Advances in Intelligent Systems and Computing, vol 1027. Springer, Singapore, 2020.

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